



TILENCORE-Gx36-Q40 CARD USER'S GUIDE

Featuring the TILE-Gx36 Processor

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Contents

CHAPTER 1 OVERVIEW

1.1 In this Release	1
1.2 About this Guide	1
1.2.1 Intended Audience	2
1.2.2 Notation Conventions	3
1.3 Technical and Customer Support	4
1.3.1 Customer Support	4
1.3.2 Product Information	4
1.3.3 Related Documents	4
1.3.3.1 Hardware Documents	4
1.3.3.2 Software Documents	4
1.4 MDE Documentation	5

CHAPTER 2 ABOUT THE TILENCORE-GX36-Q40

2.1 Product Overview	7
2.2 Purpose	7
2.3 Features	7
2.3.1 Design and Layout	7

CHAPTER 3 SYSTEM AND BOARD REQUIREMENTS

3.1 System Requirements	11
3.1.1 Hardware Requirements	11
3.1.1.1 Host System	11
3.1.1.2 Mechanical Requirements	11
3.1.1.3 Mother Board Requirements	11
3.1.1.4 Air Flow	11
3.1.2 Software Requirements	12
3.1.2.1 Operating System Requirements	12
3.1.2.2 Development Software Requirements	12
3.2 Power Requirements	12

CHAPTER 4 INSTALLING THE TILENCORE-GX36-Q40

4.1 Pre-Installation Guidelines	13
4.1.1 Handling the TILEncore-Gx36-Q40 Card	13
4.1.1.1 Preventing Electrostatic Discharge (ESD)	13
4.1.1.2 Preventing Damage to the TILEncore-Gx36-Q40 Card	13

4.2 TILEncore-Gx36-Q40 Card Installation	13
4.3 Connecting Cables	14
4.4 Installing the TILEncore-Gx36-Q40 Drivers	15

CHAPTER 5 FUNCTIONAL DESCRIPTION

5.1 TILE-Gx36 Processor Connections	17
5.1.1 TILE-Gx36 GPIO Connections	18
5.1.2 TILE-Gx36 Processor's I ² C	19
5.1.3 I ² C ROM	19
5.1.4 SPI Flash Memory	19
5.1.5 TILE-Gx36 External Interrupts	20
5.2 Power	20
5.2.1 Power Entry	20
5.2.2 Power Requirements	20
5.2.3 TILE-Gx36 Over-Temperature Protection	20
5.2.4 Core Voltage Adjustment	20
5.3 Card Resets	21
5.3.1 Power-On Reset	21
5.3.2 Push Button Reset	22
5.3.3 PCIe Resets	22
5.4 PCIe x8 Edge Finger Connector	23
5.5 DDR3 Memory Interfaces	25
5.6 QSFP	25
5.6.1 QSFP Modules	25
5.6.2 QSFP Connector	25
5.7 USB Connector	27
5.8 LEDs	27
5.8.1 Power and Status Indicators	27
5.8.2 Board Level LED Definitions	28
5.8.3 PHY LED Definitions	29
5.8.4 Interface Status Indicators	29
5.9 Heat Sinks	30
5.9.1 Passive Heat Sinks	30

CHAPTER 6 RUNNING PROGRAMS ON THE TILENCORE-GX36-Q40

6.1 Problems with Simultaneous USB and PCIe Usage	31
6.2 Boot Options	32
6.2.1 Booting a TILEncore-Gx36-Q40 via PCI	32
6.2.2 Booting a TILEncore-Gx36-Q40 via USB	33

CHAPTER 7 VERIFYING THAT THE CARD OPERATES PROPERLY

7.1 Power and Status LEDs	35
---------------------------------	----

7.2 Port Link Status/Activity LEDs 35

APPENDIX A SPECIFICATIONS

A.1 Electrical Specifications 37

A.2 Mechanical Specifications 37

A.3 Shock and Vibration Specifications 38

A.4 Environmental Specifications 38

 A.4.1 Environment 38

APPENDIX B SAFETY AND REGULATORY COMPLIANCE

B.1 Introduction 39

B.2 Safety Precautions 39

 B.2.1 Use Proper Grounding 39

 B.2.2 Use ESD Protections 39

 B.2.3 Use Caution When Working Inside a Chassis 39

 B.2.4 Avoid Fumes and Explosives 39

B.3 Regulatory Compliance 40

 B.3.1 Emissions and Susceptibility Certifications 40

 B.3.2 FCC Notice 40

 B.3.3 RoHS Statement 40

APPENDIX C TROUBLESHOOTING

APPENDIX D GLOSSARY

INDEX 47

CHAPTER 1 OVERVIEW

Thank you for purchasing the TILEncore-Gx36-Q40™ card featuring EZchip's revolutionary multicore Tile Processor™.

Note: In this document, "TILEncore-Gx36-Q40" is used to refer to various versions of the card.

1.1 In this Release

The following items, shown in [Figure 1-1](#), are included in this shipment:

- TILEncore-Gx36-Q40, featuring the TILE-Gx36™ processor. The card comes installed with the following processor:
 - TILE-Gx8036™ (Series 8000)
- Standard USB cable micro-to-A (not shown).
- Printed documentation, including the *TILEncore-Gx36-Q40 Card User's Guide* (UG421) (*this document*).



Figure 1-1: What is in the Box

Optional and Customer-Supplied Parts

- Optional cables. Customers must provide appropriate interface cables as needed.

Interface Module Requirements

If high-speed Ethernet networking is required, customer-supplied QSFP or QSFP+ modules must be installed into the TILEncore-Gx board.

1.2 About this Guide

The *TILEncore-Gx36-Q40 Card User's Guide* describes the installation, setup, and features of EZchip's TILE-Gx36™ processor. This book includes the following chapters:

- [Chapter 1: Overview](#) (*this chapter*) describes the contents of this manual.

- [Chapter 2: About the TILEncore-Gx36-Q40](#) identifies the intended user for the TILEncore-Gx36-Q40 and the card's features and supported interfaces.
- [Chapter 3: System and Board Requirements](#) specifies the power and system requirements, including the hardware and software requirements that should be followed in order to operate the card properly.
- [Chapter 4: Installing the TILEncore-Gx36-Q40](#) provides pre-installation guidelines and proper handling instructions.
- [Chapter 5: Functional Description](#) describes some of the card's features and subsystems, specifically the card resets, LEDs, interrupts, and boot options.
- [Chapter 6: Running Programs on the TILEncore-Gx36-Q40](#) provides instructions for booting the card and running a sample program.
- [Chapter 7: Verifying that the Card Operates Properly](#) describes how to test the card to verify that it has been installed properly.
- [Appendix A: Specifications](#) provides electrical, mechanical, and environmental specifications for the card.
- [Appendix B: Safety and Regulatory Compliance](#) lists the safety and compliance information for the TILEncore-Gx36-Q40.
- [Appendix C: Troubleshooting](#)
- [Glossary](#) provides definitions for terms used in this document.

An [Index](#) is also included.

1.2.1 Intended Audience

This book is intended primarily for hardware developers who want to use the TILEncore-Gx36-Q40 to jump start whole system designs, designers evaluating the TILE-Gx36™ processor for use in their boards, and individuals looking for a system solution. The TILEncore-Gx36-Q40 is offered separately from but in conjunction with the Multicore Development Environment™ (MDE), accompanying tools, software libraries, and software documentation. Contact your support engineer or sales representative for more information about the MDE.

1.2.2 Notation Conventions

Table 1-1 lists document notation conventions.

Table 1-1. Notation Conventions

Example	Description
Close command (File menu)	Titles in reference sections indicate the location of an item within the Graphical User Interface (GUI) for the MDE (for example, the Close command appears on the File menu).
Write In Progress (WIP) bit	<code>Courier</code> text indicates the names of: • A bit or bitfield, for example the <code>CHANNEL</code> bit • A special purpose register (SPR), for example the <code>RSH_COORD</code> SPR • Code sample • Application, for example the <code>tile-monitor</code> application • Command, for example the <code>link</code> command • State, for example the <code>IDLE</code> state
RSHIM MMIO registers	Blue and underlined text in <code>Courier</code> font indicates these are hypertext links to HTML files associated with the interface, or links to a specific web site.
Appendix B: Safety and Regulatory Compliance	Blue text in text font (Palatino) indicates a (cross-reference) Hypertext link to text elsewhere in the manual. If you are reading a soft-copy of this manual, you can click on the link to jump directly to the referenced section.
{this that}	Alternative required items in syntax descriptions appear within curly brackets and separated by vertical bars; read the example as this or that. One or the other is required.
[this that]	Optional items in syntax descriptions appear within brackets and separated by vertical bars; read the example as an optional this or that.
[this,...]	Optional item lists in syntax descriptions appear within brackets delimited by commas and terminated with an ellipse; read the example as an optional comma-separated list of this.
.SECTION	Commands, directives, keywords, and feature names are in text with Arial font.
<filename>	Non-keyword placeholders appear in text in italic style format and within brackets.
www.ezchip.com	Worldwide web addresses, directory names, code samples all share a <code>courier</code> font.
	Note: For correct operation, ... A Note provides supplementary information on a related topic. In the online version of this book, the word Note appears instead of this symbol.
	Caution: Incorrect device operation might result if ... Caution: Device damage might result if ... A Caution identifies conditions or inappropriate usage of the product that could lead to undesirable results or product damage. In the online version of this book, the word Caution appears instead of this symbol.

1.3 Technical and Customer Support

This section describes how to obtain additional information.

1.3.1 Customer Support

You can reach EZchip Customer Support in either of two ways:

- Visit the EZchip® Web site at
`http://www.ezchip.com/`
- E-mail questions to
`supportMC@ezchip.com`

1.3.2 Product Information

You can obtain additional product information from the EZchip Web site, from the product CD-ROM, or from EZchip printed publications.

Access the EZchip Web site online at:

`http://www.ezchip.com/`

1.3.3 Related Documents

Additional documentation is available upon request. Most of the documentation listed below is included with EZchip's MDE.

1.3.3.1 Hardware Documents

Hardware Design

- *TILE-Gx36 Processor Data Sheet (DS400)*
- *Hardware Design Guidelines for TILE-Gx Processors (WP006)*

Architecture

- *Architectural Overview for the TILE-Gx Series (UG130)*
- *TILE-Gx Instruction Set Architecture (UG401)*
- *Tile Processor and I/O Device Guide for the TILE-Gx Family of Processors (UG404)*

Boards

- *TILEncore-Gx36-Q40 Card User's Guide (UG421) (this manual)*
- *Installation Sheet for: TILEncore-Gx36-Q40 (IS117)*
- *Board Test Kit Guide for TILE-Gx Processors (UG422)*

1.3.3.2 Software Documents

Refer to [“MDE Documentation” on page 5](#).

Note: `${TILERA_ROOT}` is the root of the EZchip MDE install directory.

1.4 MDE Documentation

Note: `TILERA_ROOT` is an environment variable that points to the root of the MDE installation directory. As such it is a convenient shorthand in the documentation. It is required by many EZchip tools and examples. We recommend that you use the `tile-env` script described in *Getting Started with the Multicore Development Environment* (UG504) to set it correctly.

EZchip MDE Document Index

The EZchip MDE Document Index provides links to online copies of the MDE documentation, in both HTML and PDF formats:

```
$TILERA_ROOT/doc/index.html
```

Location of PDF Files in the Kit

The PDF versions of MDE documents can be found at this location:

```
$TILERA_ROOT/doc/pdf/
```

Man Pages and Info Pages

To view a man page for an x86 tool, run:

```
$ tile-man toolname
```

To view a man page for a Tile tool, run this on the x86 host:

```
$ tile-man toolname
```

IDE Online Help

The IDE has its own [IDE Online Documentation](#), which you can access when you use the IDE.

To get to the online help:

- Start `tile-eclipse`.

Select the menu command [Help → Help Contents](#).

- When the online help browser is displayed, select the document [IDE Online Documentation](#) in the left navigation pane, if it is not already selected.

CHAPTER 2 ABOUT THE TILENCORE-GX36-Q40

2.1 Product Overview

The TILEncore-Gx36-Q40™ PCIe card is a PCIe endpoint that enables customers to develop and test software applications on the TILE-Gx36™ processor. Physically, it is a full-height, ½-length, single slot PCI Express form factor with an 8-lane edge connector.

TILEncore-Gx36-Q40 is a modular card with a processor and two SODIMM memory modules.

2.2 Purpose

The TILEncore-Gx36-Q40 card serves three purposes: as an early development vehicle, as evaluation hardware for the multicore processor, and for integration into a “black box” system for developers who do not want to build their own hardware. Due to the wide variety of usage models, no software applications accompany the TILEncore-Gx36-Q40, but they can be obtained separately (see [“Customer Support” on page 4](#)).

2.3 Features

- TILE-Gx™ processor capable of running from 1.0GHz to 1.2GHz.
- Two modular DDR3 (SODIMM) memory interfaces.
 - DDR3 voltage is hard strapped to 1.35V. Core voltage is selectable via VIDs.
- Various interface options:
 - A single 1x40G QSFP
 - A single port 40GE FPGA-based controller
 - Standard USB accessed via micro-to-A cable (for Endpoint only)
- PCI Express standard, gold-fingered edge connector to Host, which uses a PCIe G3 physical connector.
- User-selectable boot options, including SPI Flash ROM and host boot via PCIe.

The on-board serial Boot ROM(s) are shipped with the `prebooter` (to configure the PCIe link). Initial booting is expected to be via the PCIe.
- Temperature monitoring of the environment and the TILE-Gx core.

2.3.1 Design and Layout

The TILEncore-Gx36-Q40 is comprised of two primary sections:

- The TILE-Gx processor (and its memory subsystems)
- The interface subsystem with QSFP/QSFP+ connectors

[Figure 2-1](#) and [Figure 2-3](#) depicts the major card components and associations.

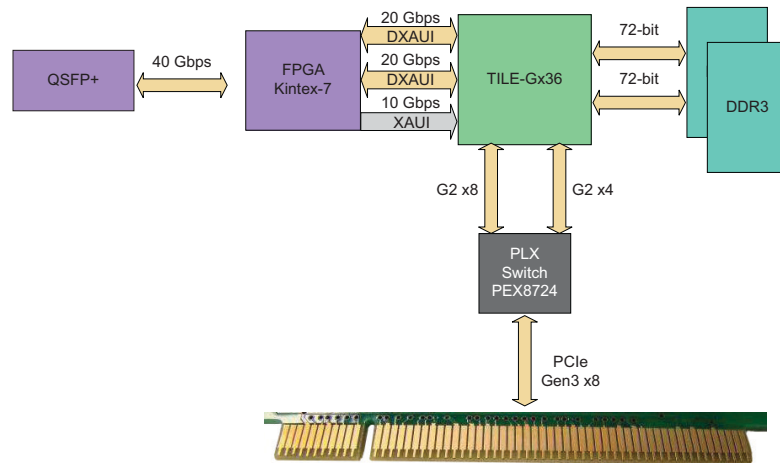


Figure 2-1: TILEncore-Gx36-Q40 Block Diagram

Figure 2-2 shows the block diagram for the FPGA, which enables the 40GbE interface for the TILEncore-Gx36-Q40 adapter. The following (optional) loopbacks, shown as light gray arrows in Figure 2-2, can be used in the chip. The Control Block (shown at the bottom) is used for setting loopback modes and getting statistics, such as link status.

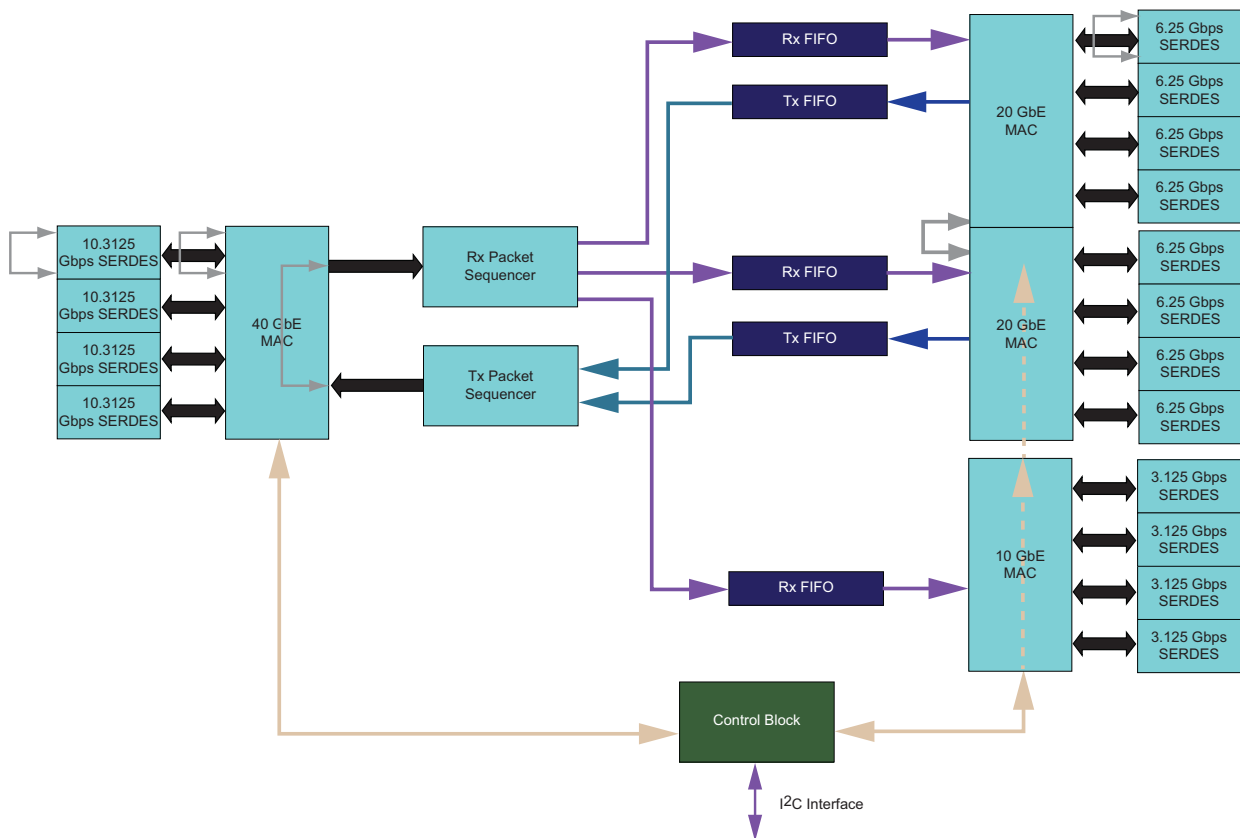


Figure 2-2: FPGA Block Diagram

Figure 2-3 shows a top view of the board and identifies the major components. For locations of the LEDs, refer to Figure 7-1.

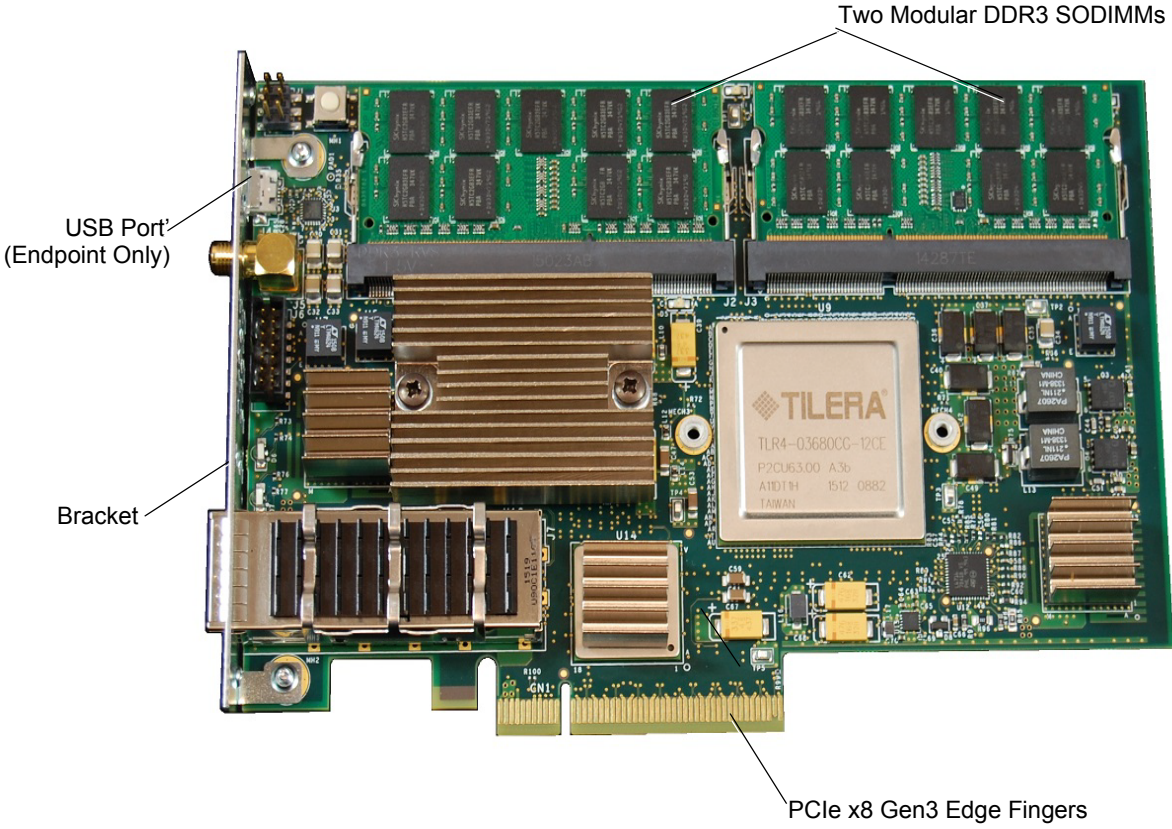


Figure 2-3: TILCore-Gx36-Q40 with QSFP/QSFP+ Interfaces

CHAPTER 3 SYSTEM AND BOARD REQUIREMENTS

3.1 System Requirements

This section lists minimal requirements for installing and using the TILEncore-Gx36-Q40™.

3.1.1 Hardware Requirements

Minimum system hardware requirements are described in the sections that follow.

3.1.1.1 Host System

EZchip® has verified that the TILEncore-Gx36-Q40 card works in many systems. For a complete list of qualified systems, refer to “PCIe Host Requirements” in the *MDE Patch Notes* (UG1118) and *MDE Release Notes* (UG908).

3.1.1.2 Mechanical Requirements

As a half-length, full-height (HF) PCIe add-in card, the TILEncore-Gx36-Q40 installs in a standard x8 or x16 PCIe slot and conforms to PCIe clearance specifications for top and bottom component heights as well as specified keepouts.

3.1.1.3 Mother Board Requirements

The TILEncore-Gx36-Q40 supports a x8 PCIe interface and requires a x8 PCIe connector. Higher bandwidth connectors are backward compatible; therefore, a x16 connector works. From an electrical standpoint, the TILEncore-Gx36-Q40 trains up to Gen3 x8 (maximum).

3.1.1.4 Air Flow

For boards installed in host systems, adhere to the following recommended operating conditions:



CAUTION!

Airflow and ambient temperature requirements:

Requires 400LFM within a fully ducted chassis.

Requires 500LFM when chassis has bypass space above the card's top edge.

Note: Do not leave the PC cover open as it can disrupt air flow inside the PC. Air circulation is best with the cover closed.

3.1.2 Software Requirements

Minimum system software requirements are described in the sections that follow.

3.1.2.1 Operating System Requirements

In order for you to operate the TILEncore-Gx36-Q40, you need the following software installed on your system:

- Operating System software should be Multicore Development Environment™ (MDE) Release 4.3.4, or later.

Refer to the *MDE Release Notes* (UG708) for specific requirements.

- Red Hat Enterprise Linux software. The MDE has been qualified on a host system running the appropriate releases of RHEL and CentOS, as specified in the MDE Release Notes (UG708) and/or MDE Patch Notes (UG718).
- EZchip includes pre-built drivers for Red Hat Enterprise Linux 5, CentOS 5, Red Hat Enterprise Linux 6, and CentOS 6. The driver source is provided; other kernels need to compile their own drivers. (EZchip provides build scripts, for example: `$TILER_ROOT/lib/modules/tilepci-compile`). Refer to MDE Release Notes (UG708) and/or MDE Patch Notes (UG718) for a list of these pre-built drivers.

The *Linux Red Hat Installation Guide* can be found at:

https://access.redhat.com/documentation/en-US/Red_Hat_Enterprise_Linux/5/html/Installation_Guide/index.html

For instructions on configuring drivers, refer to “I/O Devices and Drivers” in the *MDE System Programmer’s Guide*. For more information, contact support@ezchip.com.

3.1.2.2 Development Software Requirements

The following software is required:

EZchip Multicore Development Environment (MDE). You must use a current copy of EZchip’s MDE— at a minimum 4.3.4 revision level. This developer’s kit provides drivers and runtime code required to use the TILEncore-Gx36-Q40.

3.2 Power Requirements

The board requires 12V from the host PC power supply. It gets the +12V exclusively from the PCIe Motherboard “edge fingers” connector.

For power consumption guidelines, refer to [Section A.1 “Electrical Specifications”](#) on page 37.

Cumulative power load on the power supply must be considered when installing one or more TILEncore-Gx card(s) in a host system. Always consider the power draw required by other PCIe cards installed in the system, including the graphics card (GPU) as well.

CHAPTER 4 INSTALLING THE TILENCORE-GX36-Q40

4.1 Pre-Installation Guidelines

This section describes steps to take when installing the TILEncore-Gx36-Q40™ card.

4.1.1 Handling the TILEncore-Gx36-Q40 Card

4.1.1.1 Preventing Electrostatic Discharge (ESD)

In order to prevent ESD during shipment and storage, the TILEncore-Gx36-Q40 is placed in a static shielding bag, seated in an anti-static foam cushion, and placed in a box. This protection prevents vibration between the card and the box and avoids possible buildup of static electricity and ESD damage.

Before opening the box place it on an ESD-protected work area, preferably on a static dissipative table top. Workers should further protect the card by wearing a wrist strap that is connected to a ground point. The wrist strap is the primary means of controlling static charges on personnel. A non-conductive flooring is also recommended as walking across a conductive surface causes static electricity to build up.

When the card is not inserted in a PC, it should be kept in the protective bag in which it was shipped.

4.1.1.2 Preventing Damage to the TILEncore-Gx36-Q40 Card

When handling the TILEncore-Gx36-Q40 always hold it by the edges, not by any components, connectors or fixtures installed on the card. Do not flex the card as this can damage the solder connections.

4.2 TILEncore-Gx36-Q40 Card Installation

This section provides installation and removal guidelines only.

Note: Some motherboards have restricted slot usage. Refer to your motherboard documentation for this information.

■ Before removing or installing any cards, follow these steps:

1. Assemble the necessary tools.
2. For a standard card installation, first shut down the host system. Ensure that all power is disconnected from the PC in which it is installed.
3. Attach an antistatic wrist guard strap to a ground point.
4. Read the instructions in [“Safety and Regulatory Compliance” on page 39](#) that describe how to handle the TILEncore-Gx36-Q40 card.

■ Install a TILEncore-Gx36-Q40 card:

Follow the steps described in the Installation Sheet for the TILEncore-Gx36-Q40 Adapter card (IS117).

■ Remove a TILEncore-Gx36-Q40 card:

1. Remove the PC's cover and place it aside.

In a hot swap procedure, the host system is still be powered-on and running. Begin by deactivating the slot in which the TILEncore-Gx36-Q40 is installed. (Refer to the documentation provided with the PC chassis and/or the operating system.) Note that the green "RUN" LED on the TILEncore-Gx36-Q40 goes out, indicating that the board has been powered down and is safe to be removed. Refer to [Figure 7-1](#) for LED locations.

2. Disconnect the micro-USB cable if one is installed.
3. Remove any QSFP cartridges prior to removing the card.
4. Loosen the screws holding the card in place.

The card is held in place by two screws. Use a Phillips screwdriver (no. 2) to loosen the two screws securing the TILEncore-Gx36-Q40 card front bracket.

5. Disengage the card from the chassis, being careful to lift the card up and away from the edge connectors without twisting or flexing the card.



CAUTION! Clearance is limited between card components and the edges of the front panels of adjacent boards. Use care to avoid damaging components as you slide the card out.

6. Remove the card from the host system and place it in an antistatic bag.

4.3 Connecting Cables

EZchip provides one cable: a micro-USB cable.

■ Connect your cables:

Connect the appropriate Ethernet cables to the installed transceivers on the TILEncore-Gx36-Q40. The following cable types might be required:

Connection	Cable
QSFP w/ Cu cartridge	CAT5e or CAT6 cable.
QSFP w/ SX cartridge	1000BASE-SX OM3 Multimode Fiber (typically orange)
QSFP+ w/ SR cartridge	10GBASE-SR OM3 Multimode Fiber (typically orange)

Note: Long-range fiber connections can be used, however EZchip does not offer the QSFP/QSFP+ cartridges for this configuration.

4.4 Installing the TILEncore-Gx36-Q40 Drivers

■ Install drivers:

For instructions on installing drivers, refer to “*Installing this Release*” in the *Getting Started with the Multicore Development Environment* (UG204).

CHAPTER 5 FUNCTIONAL DESCRIPTION

This chapter describes each of the TILEncore-Gx36-Q40™ card's functional subsystems.

5.1 TILE-Gx36 Processor Connections

Table 5-1 provides a list of the Tile Processor interfaces and their connections:

Table 5-1. Tile Processor I/O Connections

Description	Interfaces to...
DDR3 Memory Interface '0'	DDR3 ECC SODIMM
DDR3 Memory Interface '1'	DDR3 ECC SODIMM
XAUI Interface '0'	FPGA (D-XAUI)
XAUI Interface '1'	FPGA (D-XAUI)
XAUI Interface '2'	FPGA (XAUI)
PCIe Lanes [0:11]	PLX PCIe Switch PEX8724
GPIO Interface	Various I/Os
UPLPI0 - USB3317C	SMC USB PHY
UART	On-board transceiver
I ² C Master 0	• DIMM SPD ROMs • BIB ROM
I ² C Master 1	FPGA and PLX
I ² C Master 2	I ² C Peripheral devices (Temp sensors and clock control)
I ² C Slave	N/A
SR0M_SPI	On-board SPI ROM and FPGA SPI ROM

5.1.1 TILE-Gx36 GPIO Connections

Table 5-2 details the connections to the GPIO pins of the Tile Processor.

Table 5-2. TILE-Gx36 Processor's GPIO Assignment

Pin Name	Usage Class	Usage Detail
GPIO[0:31]	RFU	Reserved For Future Use
GPIO[32]	GX_POWER_INT_L	From Xilinx CPLD, Power Interrupt, input, active low.
GPIO[33]	GX_TEMP_INT_L	From Xilinx CPLD, Temperature Interrupt, input, active low
GPIO[34]	PLX_SW_INTA_N	PLX Interrupt A, Input
GPIO[35]	PLX_SW_FATAL_ERR_N	PLX Fatal Interrupt, Input
GPIO[36]	USB_OC_FLAG_L	USB Over Current FLAG_L, input, active low
GPIO[37]	RFU	Reserved For Future Use
GPIO[38]	GX_SMA_PULSE	Bracket SMA Pulse, Input
GPIO[39]	GX_SMA_FPGA_GPIO	SMA Pulse GPIO Control, Output
GPIO[40]	GX_USB_RESET_L	Reset to CPLD for USB Controller, output, active low.
GPIO[41]	GX_PCIE_PERST_N	Reset to CPLD for PLX Switch, output, active low.
GPIO[42]	GX_FPGA_RESET_L	Reset to CPLD for FPGA, output, active low.
GPIO[43]	RFU	Reserved For Future Use
GPIO[44]	RFU	Reserved For Future Use
GPIO[45]	GX_FPGA_INIT_B	Initialize Command to FPGA, output, active low.
GPIO[46]	GX_FPGA_PROG_L	Program Command to FPGA, output, active low.
GPIO[47]	GX_SROM_EN	Program Enable to FPGA SPI ROM, output, active low
GPIO[48:63]	RFU	Reserved For Future Use

5.1.2 TILE-Gx36 Processor's I²C

There are currently three master interfaces connected to the TILE-Gx36 Processor's I²C Master interface directly mounted on the TILEncore-Gx36-Q40 card. There are currently nine devices connected to the TILE-Gx's I²C Master Interface directly mounted on the TILEncore-Gx36-Q40 card. [Table 5-3](#) lists the device address, part type, and description.

Note: While there is one slave interface on the TILE-Gx36 processors' I²C slave interface, the slave interface is unused in the TILEncore-Gx36-Q40 design.

Table 5-3. I²C Address and Device Map

I ² C Address 7-bit Address	Master Interface Number	Vendor and Part Number	Description
1010 100X	I2CM0	Atmel - AT24C512	2-wire ROM, 512 Kbits (Atmel), Boot ROM 0
1010 000X	I2CM0	Atmel - type ROM	DDR3 I/F 0 SPD ROM - DIMM
1010 001X	I2CM0	Atmel - type ROM	DDR3 I/F 1 SPD ROM - DIMM
0111 000X	I2CM1	PLX - PEX8724	PLX Switch
0111 1000	I2CM1	Xilinx FPGA	XC7J325T-FFG900
0011 000X	I2CM2	National Semi - LM95235QQ	TILE-Gx36 Temperature Sensor
1001 100X	I2CM2	National Semi - LM95235QQ	PLX Temperature Sensor
0101 001X	I2CM2	National Semi - LM95235QQ	FPGA Temperature Sensor
1111 010X	I2CM2	IDT - IDT8T49N281I	Clock Driver Chip

5.1.3 I²C ROM

An I²C 2-wire serial EEPROM is available for user configuration data. It is connected to the TILE-Gx Processor's I2CM0 interface. The device is an Atmel AT24512 or equivalent 512K part.

5.1.4 SPI Flash Memory

In addition to the I²C 4-wire serial EEPROM there is a SPI ROM for boot loading, or other user configuration data. The SPI ROM also includes the PCIe prebooter, which contains system configuration data. The SPI ROM is connected to the TILE-Gx36's SPI interface. The device is a Numonyx N25Q128, 128Mbit part.

5.1.5 TILE-Gx36 External Interrupts

Table 5-4 details the interrupts to the TILE-Gx36.

Table 5-4. TILE-Gx Processor's Interrupt Assignment

Interrupt Name	Interrupt Source	Physical Connection
GX_POWER_INT_L	Xilinx CPLD	TILE-Gx36 GPIO00[32]
GX_TEMP_INT_L	LM95235Q Temperature Sensor. When asserted LOE provides an early warning that board power is going down due to over temperature.	TILE-Gx36 GPIO00[33]
PLX_SW_INTA_N	PLX PCIe Switch	Gx36 GPIO00[34]
PLX_SW_FATAL_ERR_N	PLX PCIe Switch	Gx36 GPIO00[35]

5.2 Power

5.2.1 Power Entry

Power is supplied from the PCIe edge connector coupled to the power supply in the chassis. All other sub-voltages are generated by switching power supplies or low-dropout (LDOs) regulators on the board.

5.2.2 Power Requirements

Only +12V power, with a regulation tolerance equal to +/-5% (11.4V - 12.6V), is required from the connector. The output ripple must equal 120 mV pp.

Note: The PCIe card does not tolerate “sag” or transients outside of these tolerances.

5.2.3 TILE-Gx36 Over-Temperature Protection

The National Semiconductor LM95235Q temperature sensor is used to protect the TILE-Gx36 die from going above a programmed temperature threshold. Active LOW output travels to two locations:

- GPIO[33] of the TILE-Gx36. Software uses this as an early-warning indicator that the card power is going to shut off soon, in case any process clean-up or file-logging needs to be done.
- A power sequencer that issues an overtemp event puts the sequencer into reset. All voltage rails are disabled, which shuts off the board.

5.2.4 Core Voltage Adjustment

The board dynamically adjusts its core voltage to compensate for a series of factors, like variations in the manufacturing (silicon) process. A set of VID_CORE output pins in the Tile Processor aids in this process. The TILE-Gx processors are designed to operate with a VR11.1-compliant variable output DC regulator delivering the core supply voltage (VDD).

Upon power-up, the pins default to a setting for 0.95V (0x6A). During the early stages of the boot process and prior to the core clock PLL moving to its final speed, the VID_CORE pins are reconfigured to their final setting. A subsequent hard reset restores the VID_CORE pins to their default settings. The process repeats each time the chip reboots.

For more information refer to the VID tables in the *TILE-Gx36 Processor Data Sheet* (DS400).

5.3 Card Resets

The card supports several reset options and each reset source can generate a system reset, which asserts a hard reset to the TILE-Gx36 processor. All other devices are under software control of the Tile processor, so a hard reset of the TILE-Gx36 processor propagates to all the other resettable devices on the board including the XGbE PHY, USB PHY, and PCIe endpoint. See [Table 5-5](#) for a reset propagation matrix and the subsections below for further information regarding specific device and interface reset conditions.

Note: All reset lines are active LOW.

Table 5-5. Reset Propagation

Reset Source	TILE-Gx36 Reset	Description
Power-On	Yes	Controlled by on-board power sequencer/monitor.
Push Button Reset	Yes	Controlled by push button switch (SW1).
PCIe Reset	Yes	The card is configured as an end point (add-in card) and reset is generated by the host.

5.3.1 Power-On Reset

Power-On reset is controlled by the voltage monitor/sequencer (the Texas Instruments UCD9081), which senses when +12V has been applied and enables the switching regulators. When all the voltages are between -6.5% and +25% of their nominal voltage, the power-on reset signal is de-asserted.

The 12V supply is monitored and sequenced by the Linear Tech LTC4211 Hot Swap controller, which is (by default) enabled by the presence of +12V on the PCIe edge connectors. The Hot Swap controller fuses are set to 8.3A and ramp the 12V. Once +12V reaches 79% of its rise, the LTC4211 enables the UCD9081 to sequence up the remaining board regulators.

The sections that follow provide information regarding specific device and interface reset conditions.

5.3.2 Push Button Reset

The push button reset switch (SW1) can be used to reset the TILEncore-Gx36-Q40.

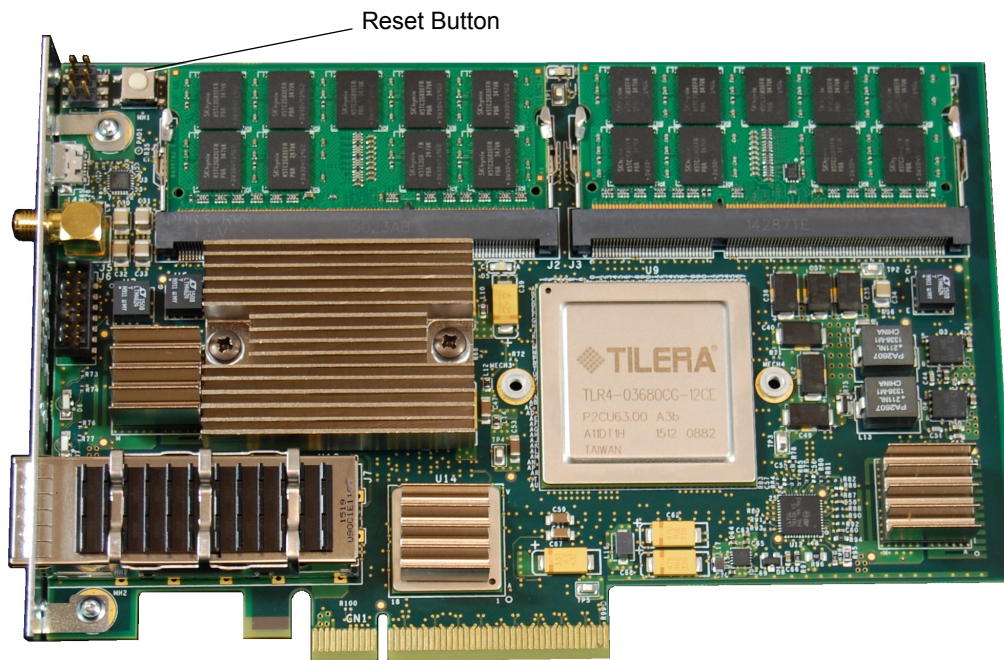


Figure 5-1: Location of Push Button Reset

5.3.3 PCIe Resets

PCIe fundamental reset connects to the TILEncore-Gx36-Q40 card from the PCIe edge fingers, and is distributed from an onboard CPLD to the following endpoints: Gx36 and the PLX PCIe Switch.

The TILE-Gx36 is always a PCIe Endpoint, and the PCIe reset will be logically connected to the TILE-Gx36's hard reset signal, which causes the TILE-Gx36 to execute a Level 0 boot.

5.4 PCIe x8 Edge Finger Connector

Table 5-6. PCIe X8 Edge Finger Connector Summary

Pin	A Side Connector		B Side Connector	
#	Board Signal Name	PCIe Description	Board Signal Name	PCIe Description
1	PCIE_PRSENT_L	PRESENT#1	EDGE_FINGER_12V	+12v
2	EDGE_FINGER_12V	+12V	EDGE_FINGER_12V	+12v
3	EDGE_FINGER_12V	+12V	EDGE_FINGER_12V	+12v
4	GND	GROUND	GND	GROUND
5	NC	JTAG BUS TCK	NC	SMB BUS CLK
6	PCIE_TDI	JTAG3_TDI	NC	SMB BUS DATA
7	PCIE_TDO	JTAG4_TDO	GND	GROUND
8	JTAG5_TMS	JTAG BUS	PM_3_3v	PCIe MEZZ 3.3v
9	PM_3_3v	PCIe MEZZ 3.3v	NC	JTAG TRST_L
10	PM_3_3v	PCIe MEZZ 3.3v	NC	3.3VAUX
11	PERST#	PERST#	NC	WAKE#
12	GND	GROUND	NC	RSVD3
13	PCIE_REFCLK_EDGE_P	REFCLK+	GND	GROUND
14	PCIE_REFCLK_EDGE_N	REFCLK-	PCIE_RX0_P	PETp0
15	GND	GROUND	PCIE_RX0_N	PETn0
16	PCIE_TX0_P	PERp0	GND	GROUND
17	PCIE_TX0_N	PERn0	NC	PRSNT2X1#
18	GND	GROUND	GND	GROUND
19	NC	RSVD1	PCIE_RX1_P	PETp1
20	GND	GROUND	PCIE_RX1_N	PETn1
21	PCIE_TX1_P	PERp1	GND	GROUND
22	PCIE_TX1_N	PERn1	GND	GROUND
23	GND	GROUND	PCIE_RX2_P	PETp2
24	GND	GROUND	PCIE_RX2_N	PETn2

Table 5-6. PCIe X8 Edge Finger Connector Summary (continued)

Pin	A Side Connector		B Side Connector	
#	Board Signal Name	PCIe Description	Board Signal Name	PCIe Description
25	PCIE_TX2_P	PERp2	GND	GROUND
26	PCIE_TX2_N	PERn2	GND	GROUND
27	GND	GROUND	PCIE_RX3_P	PETp3
28	GND	GROUND	PCIE_RX3_N	PETn3
29	PCIE_TX3_P	PERp3	GND	GROUND
30	PCIE_TX3_N	PERn3	NC	RSVD4
31	GND	GROUND	PCIE_PRSNT2X4_L	PRSNT2X4#
32	NC	RSVD2	GND	GROUND
33	NC	RSVD5	PCIE_RX4_P	PETp4
34	GND	GROUND	PCIE_RX4_N	PETn4
35	PCIE_TX4_P	PERp4	GND	GROUND
36	PCIE_TX4_N	PERn4	GND	GROUND
37	GND	GROUND	PCIE_RX5_P	PETp
38	GND	GROUND	PCIE_RX5_N	PETn
39	PCIE_TX5_P	PERp5	GND	GROUND
40	PCIE_TX5_N	PERn5	GND	GROUND
41	GND	GROUND	PCIE_RX6_P	PETp
42	GND	GROUND	PCIE_RX6_N	PETn
43	PCIE_TX6_P	PERp6	GND	GROUND
44	PCIE_TX6_N	PERn6	GND	GROUND
45	GND	GROUND	PCIE_RX7_P	PETp
46	GND	GROUND	PCIE_RX7_N	PETn
47	PCIE_TX7_P	PERp7	GND	GROUND
48	PCIE_TX7_N	PERn7	PCIE_PRSNT2X8_L	PRSNT2X8#
49	GND	GROUND	GND	GROUND

5.5 DDR3 Memory Interfaces

The two memory interfaces of the TILE-Gx36 are each connected to a SODIMM socket for easy user modification of speed and density. Though we have a target configuration of 4Gbyte/interface density, component availability or customer requirements make these interfaces a “build option”. Configuration data for these memory interfaces is stored in the SODIMM SPD (2-wire serial) ROM so that the TILE-Gx36 software can configure its memory controllers to work correctly with the memory.

Note: Only ECC SODIMMs should be used in the TILEncore-Gx36-Q40.

5.6 QSFP

5.6.1 QSFP Modules

The following modules have been approved for use with the TILEncore-Gx36-Q40: TBD

5.6.2 QSFP Connector

The QSFP Connectors are made up of two parts: the SMT Connector and the metal enclosure that surrounds it.

Table 5-7. QSFP Connector Summary

Pin	Signal Name	Description
1	GND	Ground
2	Tx2n	Transmitter Inverted Data Input
3	Tx2p	Transmitter Non-Inverted Data Input
4	GND	Ground
5	Tx4n	Transmitter Inverted Data Input
6	Tx4p	Transmitter Non-Inverted Data Input
7	GND	Ground
8	ModSelL	Module Select
9	ResetL	Module Reset
10	Vcc Rx	+3.3V Power Supply Receiver
11	SCL	2-wire serial interface clock
12	SDA	2-wire serial interface data
13	GND	Ground
14	Rx3p	Receiver Non-Inverted Data Output
15	Rx3n	Receiver Inverted Data Output

Table 5-7. QSFPP Connector Summary (continued)

Pin	Signal Name	Description
16	GND	Ground
17	Rx1p	Receiver Non-Inverted Data Output
18	Rx1n	Receiver Inverted Data Output
19	GND	Ground
20	GND	Ground
21	Rx2n	Receiver Inverted Data Output
22	Rx2p	Receiver Non-Inverted Data Output
23	GND	Ground
24	Rx4n	Receiver Inverted Data Output
25	Rx4p	Receiver Non-Inverted Data Output
26	GND	Ground
27	ModPrsL	Module Present
28	IntL	Interrupt
29	Vcc Tx	+3.3V Power supply transmitter
30	Vcc1	+3.3V Power supply
31	LPMode	Low Power Mode
32	GND	Ground
33	Tx3p	Transmitter Non-Inverted Data Input
34	Tx3n	Transmitter Inverted Data Input
35	GND	Ground
36	Tx1p	Transmitter Non-Inverted Data Input
37	Tx1n	Transmitter Inverted Data Input
38	GND	Ground

5.7 USB Connector

Connector Details

- USB Connector, MICRO USB AB RECPT RA SMT BTM MNT ASSY
- Molex P/N: 47589-0001

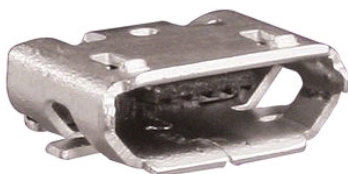


Figure 5-2: Micro USB Connector

Table 5-8. Micro USB Connector Summary

Pin	Signal Name	Description	Class
1	USB_VBUS	V Bus	N/A
2	USB_DM	USB Data Minus	N/A
3	USB_DP	USB Data Plus	N/A
4	USB_ID_IN	USB ID	N/A
5	GND	Power Return	N/A

5.8 LEDs

5.8.1 Power and Status Indicators

The main board features seven LEDs in total on TILEncore-Gx36-Q40. Three power related LEDs, one TILE-Gx36 related, and one FPGA program related, as listed in [Table 5-9](#). Two are Port Link Status/Activity LEDs that are controlled by the FPGA/PHY, as listed in [Table 5-10](#).

[Figure 5-3](#) shows the location LEDs on the TOP side of the board. Note that this does not include Port Link Status/Activity LEDs that are controlled by the PHY.

For more information on Link Status LEDs, refer to [Figure 5-3](#).

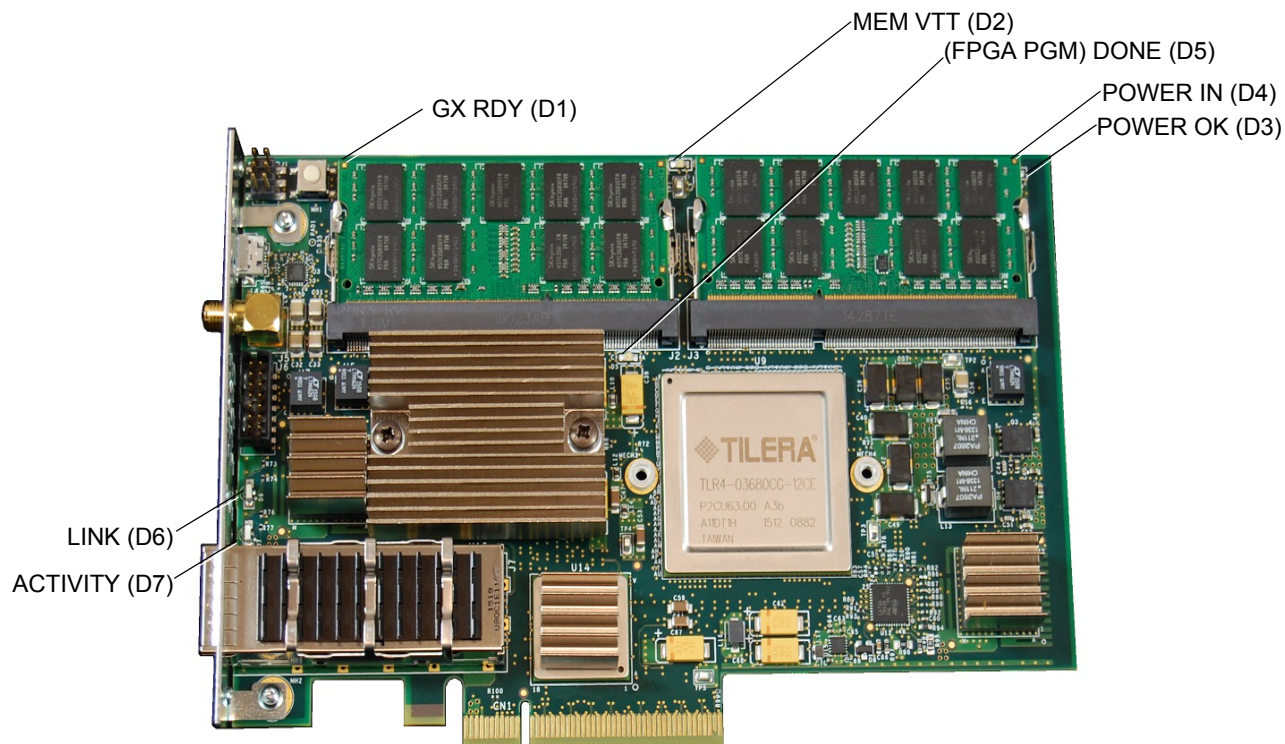


Figure 5-3: TILEcore-Gx36-Q40 Showing the Locations of LEDs

5.8.2 Board Level LED Definitions

Table 5-9. Board-Level LED Definitions

Reference	Color	Label	Location	Description
D1	GREEN	GX READY	Back	This LED indicates that the core RefClk is present, reset has been removed, and the internal configuration is complete.
D2	GREEN	MEM VTT	Front	This LED is tied to MEM_VTT to indicate that Memory Termination power supply is healthy.
D3	GREEN	POWER OK	Front	This LED is tied to 3.3V to indicate that all secondary voltages are up and running on the card.
D4	GREEN	POWER IN	Front	The hot swap controller is enabled and +12 V is being provided to the card's on-board regulators.
D5	GREEN	(FPGA PGM) DONE	Front	This LED is tied to FPGA DONE to indicate that the FPGA is DONE loading its program.

5.8.3 PHY LED Definitions

Table 5-10. PHY LED Definitions

Reference	Color	Label	Location	Description
D6	GREEN	LINK	Bracket	Indicates Network Link Partner.
D7	YELLOW	ACTIVITY	Bracket	Indicates Network Link Activity.

The Power LED, located on the Rear Plate glows green only when all components are powered on (see [Figure 5-4](#)).

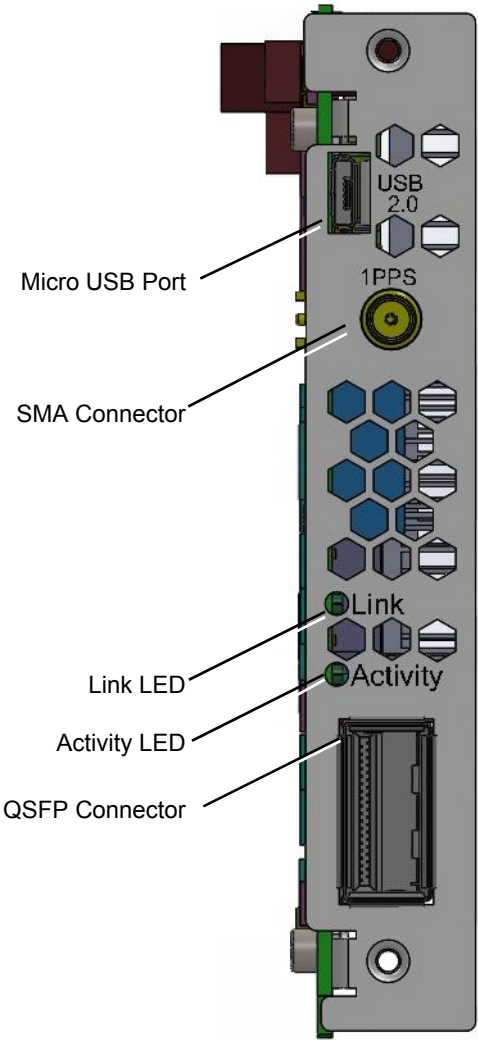


Figure 5-4: Link LEDs

5.8.4 Interface Status Indicators

Status indicators on the rear plate show the state of the Ethernet links.

A pair of LED indicators is located next to each port. The green indicator is for link status (up or down) and the yellow indicator is for activity.

5.9 Heat Sinks

5.9.1 Passive Heat Sinks

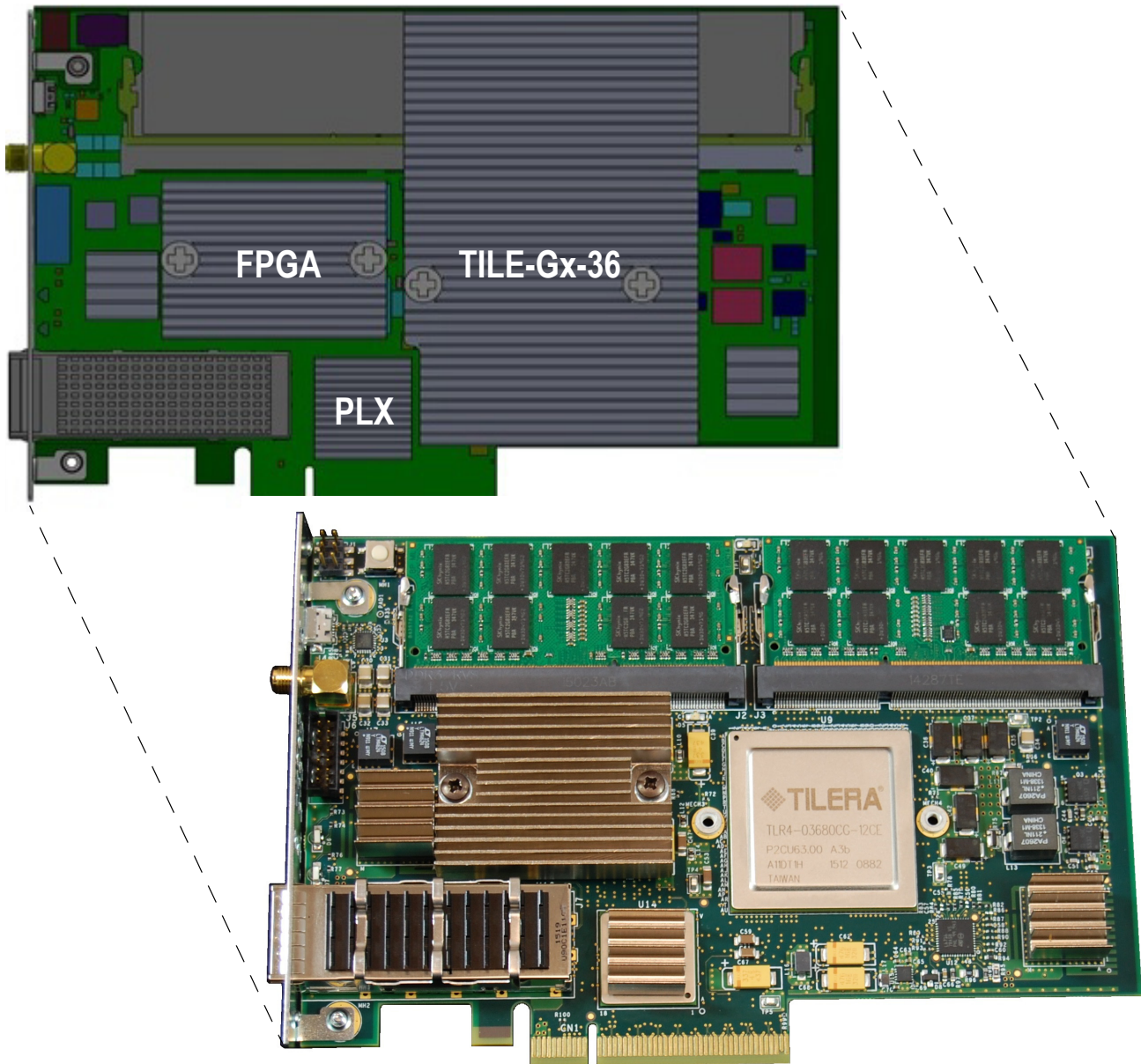


Figure 5-5: Passive Heat Sinks — Location

CHAPTER 6 RUNNING PROGRAMS ON THE TILENCORE-GX36-Q40

6.1 Problems with Simultaneous USB and PCIe Usage

The TILEncore-Gx™ card supports boot and debug operations via its PCIe interface. It is also possible, although not recommended, to boot and debug the card via its USB device interface. You can connect both of these interfaces simultaneously, but doing so can cause undesired system behavior. This section explains why that is and how to avoid it.

The problem is related to booting the TILE-Gx™ processor, which happens whenever you run `tile-monitor` *without* the `--resume` option, or whenever you run any chip tests via the Board Test Kit (BTK). The first step of booting the TILE-Gx processor is to reset it. This operation is done automatically by the `tileusb` or `tilegxpci` drivers when you start to boot the processor via the corresponding interfaces.

When the processor is reset, all of the I/O interfaces on the chip get reset. In the case of USB, the USB device appears to be disconnected from the host; in the case of PCIe, the PCIe link between the host and the processor goes down. Soon afterward, communication is re-established.

In the case of USB, the link is brought back up automatically by the processor; in the case of PCIe, the PCIe prebooter is loaded from the SROM and brings up the link. However, the host's response to these momentary interruptions is very different. USB is designed as a hot-pluggable interface. The cabling for USB devices is generally accessible to their users, and many of those devices are connected to their host system only temporarily (as when you download photographs from a digital camera). Because of this, the USB software stack expects devices to be disconnected and reconnected.

While PCIe devices can also be hot-pluggable, this is the exception, rather than the rule. Desktop computers, which are commonly used to host TILEncore-Gx36-Q40 cards, typically do not support hot-plugging. Thus, their PCIe software stacks do not expect PCIe devices to be disconnected.

In order to cope with the fact that rebooting (and thus resetting) a TILE-Gx36 processor causes the USB or PCIe connection to be interrupted, the MDE's `tileusb` and `tilegxpci` drivers perform special steps as part of a reboot operation. The driver might need to read the configuration state from the device, though state information is destroyed by the reset. Once the device has been reconnected, the configuration state is restored. The driver might also need to reconfigure the system temporarily to tolerate the upcoming disconnect event. However, these steps can only be performed if the relevant driver knows that a reset is about to happen, which brings us to the problem with simultaneous USB and PCIe connections.

Imagine what happens when a TILEncore-Gx card is configured into a host system and both its USB and PCIe ports are connected. Both the `tileusb` and `tilegxpci` drivers believe they are responsible for controlling the card, and both devices are willing to reboot and thus reset it. However, neither driver has any "knowledge" of the other. Thus, if you use the `tileusb` driver to

reset the card, the `tilegxpci` driver does not get the opportunity to do any of the normal state-saving and disconnection preparations; the host system sees an unexpected interruption of service to one of its PCIe devices instead.

The effect of this kind of event is system-dependent. On some systems, the unexpected PCIe disconnect leads to a fatal error and a system crash. On others, the system does not crash, but the fact that the newly-reset TILEncore card has lost the configuration information programmed into it during device discovery can lead to problems with PCIe I/O, perhaps not even involving the TILEncore card. In either case, the only way to recover is to reboot the host system.

Because of this, we strongly recommend that if you have a TILEncore-Gx36 connected via both PCIe and USB, that you never reboot the card via USB.

This means that you should not run `tile-monitor` against the USB interface, and should not run any tests when using `tile-btk` against the USB interface. Rebooting such a doubly-connected card via PCIe, on the other hand, is generally safe, since the USB interface is much more tolerant of momentary interruptions.



If you have PCIe and USB connected at the same time, **you should avoid resetting the card via USB.**

If you reset the card over USB, one of two things is going to happen:

- If your card is in a host **that is not sensitive to the PCI link going down**, the PCI link goes down and comes back up, but the chip is not programmed properly and you are be able to talk to the card over PCIe. It is also possible that the default settings on the card might interfere with other things on the BUS, which is less likely but still possible. The main symptom is that you are not able to talk to the card over PCIe. The only way to fix that problem is to reboot the host system.
- If your card is in a host **that is sensitive to the link going down**, the system most likely crashes and reboots. Some hosts might not be set up to reboot after a failure like that. You might need to power cycle the host in these cases.

6.2 Boot Options

The TILEncore-Gx36-Q40 card supports the following boot options:

- Boot from host via the TILEncore-Gx36-Q40™ card's PCIe 0 or 1 Interface.

This is the standard boot option.

- Boot from boot ROM (SPI ROM).

Requires programming of SPI ROM.

- Boot from USB. As mentioned in [Section 6.1](#), although this feature is supported, booting from USB is not recommended.

6.2.1 Booting a TILEncore-Gx36-Q40 via PCI

```
tile-monitor --dev gxpci0 \
--hvx TLR_NETWORK=none --hvd POST=quick \
--hvx "dataplane=1-35"
```

6.2.2 Booting a TILEncore-Gx36-Q40 via USB

To run a program on the TILEncore-Gx36-Q40™ card for exhibition purposes, use the `--dev` option:

```
tile-monitor --dev usb0 \  
--hvd POST=quick \  
--hvx TLR_NETWORK=none \  
--hvx "dataplane=1-35"
```

You can also run against an already-booted TILEncore-Gx36-Q40 card without rebooting it by using the `--resume` option:

```
tile-monitor --dev usb0 --resume
```


CHAPTER 7 VERIFYING THAT THE CARD OPERATES PROPERLY

7.1 Power and Status LEDs

The TILEncore-Gx36-Q40™ card has four Power and Health LED indicators located on top of the card. These are defined in [Table 5-9 on page 28](#). When a card is in good working order, powered up, and ready for operation, all four GREEN LEDs (D4, D3, D5, D2 and D1) should be lit.

For additional information, refer to [Section 5.8 “LEDs”](#) on page 27.

7.2 Port Link Status/Activity LEDs

The LINK LED, located on the bracket and depicted in [Figure 7-1](#), indicates that the Physical Link is operating properly.

■ To test the TILEncore-Gx36-Q40:

The quick test, `tile-btk`, can be run to test the chip’s internal data paths, basic peripherals, XAUI Internal loopback, and memory. Run `tile-btk` from a terminal window.

Note: The user interface of the test kit should be invoked as follows:

```
$TILER_ROOT/bin/tile-btk
```

For more information about using the Board Test Kit (BTK), refer to *Board Test Kit Guide for TILE-Gx Processors* (UG422).

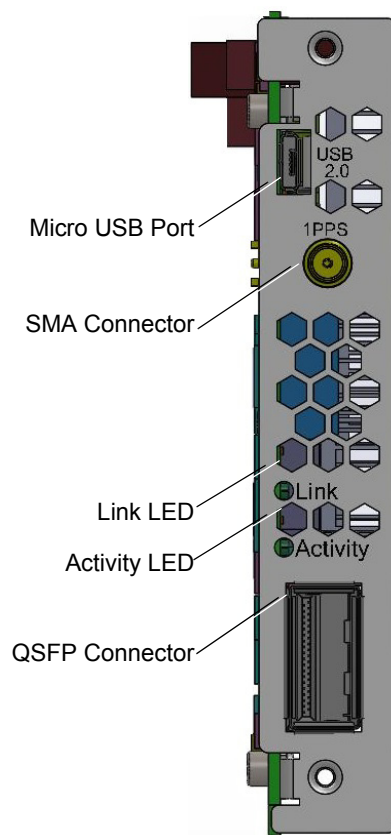


Figure 7-1: QSFP/QSFP+ Link and Power LEDs

APPENDIX A: SPECIFICATIONS

A.1 Electrical Specifications

Power consumption varies depending on application, but [Table A-1](#) can be used as a guide.

Note: Power consumption data has not been verified under live testing conditions. The information provided is subject to change.

Table A-1. Power Consumption

Card Version	Typical
TILEncore-Gx36-Q40 card @ 900MHz	70W (5.8A @ 12V) ^a
TILEncore-Gx36-Q40 card @ 1.0GHz	75W (6.25 @ 12V) ^a

a. Wattage based on the application used, temperature, and network loading.

See also [Section 3.2 “Power Requirements”](#) on page 12.

Note: The board consumes no more than ~66 watts.

A.2 Mechanical Specifications

The mechanical dimensions of the card are compliant with the add-in card form factor for an 8-lane PCI Express “short” add-in card.

Table A-2. Physical Specifications

Board Height	11.1 cm (4.38 in)
Board Length	16.8 cm (6.6 in)
Bracket Height	12 cm (4.72 in)

A.3 Shock and Vibration Specifications

The TILEncore-Gx36 is built to withstand shock and vibration that would be typical in shipping and in non-mobile operating environments. If the board is to be shipped while installed in a host server, it is recommended that the rear bracket be firmly secured into the chassis and that the opposite edge of the card (with the DC power connector) be well supported.

Table A-3 lists the shock and vibration test specifications TILEncore-Gx36-Q40 card.

Table A-3. Table A-3. Shock and Vibration Specifications

Specification	Range
Random Vibration	5 - 500Hz @ 1G rms (9.81 m/s ² rms), X, Y, and Z axis
Shock	Half-Sine @ 80G (785 m/s ²), 2 msec duration, +/- X, Y, Z axis

Note: Detailed test reports can be provided upon request.

A.4 Environmental Specifications

A.4.1 Environment

Table A-4 lists the other environmental specifications of the TILEncore-Gx36-Q40 card.

Table A-4. Chassis-Level Environmental Specifications

Specification	Range
Storage temperature	-40° C to +70° C
Operating temperature	0° C to +45° C
Operating humidity	5% to 95%, non-condensing
Vibration	5 Hz to 20 Hz: 0.01 g ² /Hz sloping up to 0.02 g ² /Hz
	20 Hz to 500 Hz: 0.02 g ² /Hz (flat)
Air Flow	Airflow and ambient temperature requirements: - Requires 400LFM within a fully ducted chassis. - Requires 500LFM when chassis has bypass space above the card's top edge.

APPENDIX B: SAFETY AND REGULATORY COMPLIANCE

B.1 Introduction

This chapter explains important safety precautions that you should observe when operating the TILEncore-Gx36-Q40™ card.

Note: Follow all warnings and instructions marked on the equipment.

B.2 Safety Precautions

The TILEncore-Gx36-Q40 card is designed and tested to the following safety requirements:

- UL 60950-1 Safety
- EN 60950 IEC 60950-1
- CSA C22.2 No. 60950-1

B.2.1 Use Proper Grounding

To avoid electric shock, make sure that the system is connected to an electrical safety (green wire) ground.

B.2.2 Use ESD Protections

The TILEncore-Gx36-Q40 contains static-sensitive devices. Static control procedures should be used when installing, removing, and handling all components. Use an antistatic wrist strap at all times. Wrist straps usually come with an alligator clip that you clip to the metal frame of the chassis after the cover is removed.

Never touch the card with any conductive objects.

For additional handling instructions, refer to [Section 4.1.1 “Handling the TILEncore-Gx36-Q40 Card”](#) on page 13.

B.2.3 Use Caution When Working Inside a Chassis



Use caution when working inside the computer chassis as touching or bumping the card's rotating blower module can damage it.

B.2.4 Avoid Fumes and Explosives

Do not operate the TILEncore-Gx36-Q40 card in the presence of explosives or inflammable gases or fumes.

B.3 Regulatory Compliance

B.3.1 Emissions and Susceptibility Certifications

The TILEncore-Gx36-Q40 card is designed and tested for EMI emissions and susceptibility to the following standards:

- FCC part 15, subpart B, Class A (commercial)
- CE EN 55022: 1998 Class A
 - Conducted emissions
 - Radiated emissions
- CE EN 55024: 1998
 - Immunity for ITE Amendment A1: 2001
- CE EN 61000-3-2 2000
 - Harmonic current emissions
- CE EN 61000-3-3
 - Voltage fluctuations and flicker
- CE IEC 6100-4-2 1995
 - ESD Air Discharge 8kV Contact discharge 4kV
- CE IEC 6100-4-3 1995
 - Radiated Immunity (80 - 1000MHz), 3V/m 80% AM by 1kHz
- CE IEC 6100-4-4 1995
 - EFT/B; Immunity to electrical fast transients; 1kV Power leads
- CE IEC 6100-4-5 1995
 - Immunity to conducted surges; common mode; 2kV
- CE IEC 6100-4-6 1996
 - Conducted immunity
- CE IEC 6100-4-11 1994
 - Voltage dips and short interruptions

B.3.2 FCC Notice

This equipment generates, uses, and can radiate radio frequency energy. If the card is not installed correctly and used in accordance with the instruction manual, it can cause interference to radio communications.

B.3.3 RoHS Statement

The TILEncore-Gx36-Q40 is built to comply with the 2002/95/EC European legislation of 2003-02-13, which states that equipment must be free of:

- Lead
- Mercury
- Cadmium

- Hexavalent Chromium
- Polybrominated Biphenyls (PBB)
- Polybrominated Diphenyl Ethers (PBDE)

Products free of the materials are known as being RoHS-6 compliant. However, the TILE-Gx process has a small amount of lead contained in the internal die bond for the “flip chip” package. This means that at the board level, this product is specified to be RoHS-5 compliant. However, there are no other devices containing lead on the PCB, and the soldering technology for all components is lead-free.

APPENDIX C: TROUBLESHOOTING

This chapter describes basic methods for troubleshooting your TILEncore-Gx36-Q40™ card.

How do I install the PCIe prebooter?

Boot the board using USB as described in [Section 6.2 “Boot Options”](#) on page 32. Connect a serial console device to the UART.

From the console port, type:

```
sbim --info
sbim --install-prebooter /boot/pci-preboot.bin
```

At the prompt, respond with “Yes” and then enter:

```
sbim --info
```

The system responds with something like:

```
SROM booter: Present 8336 bytes
Time Installed: Tue
```

What does tile-console do?

The `tile-console` command provides access to the console port on a Tile Processor, which displays Linux boot messages and normally provides an interactive Linux login, via a serial device on the host.

Use the same device name nomenclature as with `tile-monitor`. For example:

```
--dev usb0
—OR—
--dev gxpci0
```

With tile-btk operations, I cannot reboot the TILEncore-Gx36-Q40 card with `--dev gxpci0` after boot it with `--dev usb0`. Is this expected?

Yes, you can expect this behavior. We recommend power-cycling the host *after* every time you run the btk commands.

Even if the btk can reboot the TILE-Gx from the SROM and potentially get the PCIe link up, the host does not assign any BAR space to the TILE-Gx, unless we support PCI re-scan, which we do not. Therefore, no further PCI communication can happen.

APPENDIX D: GLOSSARY

Term	Definition
2-WIRE	2-Wire Serial Clock and Data (SCL and SDA) operating up to 400kHz.
4-WIRE	4-WIRE, or two pair, Synchronous Serial Interface or Serial Peripheral Interface (SPI).
BIB	Board Information Block. The EZchip® run-time software stack uses a set of system-specific information to govern the configuration and initialization of the Tile Processor™ and the devices to which it connects. This data is contained in a structure called the Board Information Block (BIB).
CPLD	Complex Programmable Logic Device
DDR3	Double Data Rate, Type 3. Synchronous dynamic random access memory—a modern kind of dynamic random access memory (DRAM) with a high bandwidth interface.
DIP	Dual In-line Package.
DIMM	Dual In-line Memory Module • LRDIMM – Load reduced • RDIMM – Registered • UDIMM - Unbuffered
ECC SODIMM	Error Check and Correction Small Outline DIMM.
EEPROM	Electrically Erasable Programmable Read-Only Memory.
Gbps	Gigabit per second.
GBps	Gigabytes per second.
GbE	Gigabit Ethernet.
GPIO	General Purpose I/O.
I/O	Input/Output.
JTAG	Joint Test Action Group
LFM	Linear Feet per Minute
LED	Light-Emitting-Diode used for status indicator
LVPECL	Low-Voltage Positive-referenced Emitter-Coupled Logic
MDE™	Multicore Development Environment™. EZchip's multicore programming environment.
PCI	Peripheral Component Interconnect, a bus standard for connecting peripheral devices to a computer motherboard.

Term	Definition
PCIe or PCI-e	PCI Express, a PCI interface format using high-speed serial links for the connection between peripheral devices and the motherboard.
POR	Power-On Reset.
QSFP	Quad Small Form-factor Pluggable.
ROM	Read-Only Memory.
SODIMM	Small Outline DIMM.
SPD ROM	Serial Presence Detect ROM.
SPI	Serial Peripheral Interface.
SPI-SROM	Serial Flash with serial peripheral interface.
USB	Universal Serial Bus
VID	Voltage Identification (bits)
XAUI	10 Gigabit Attachment Unit Interface, a standard for connecting 10 Gigabit Ethernet.
XGbE	10 Gigabit Ethernet

I INDEX

Symbols

(FPGA PGM) DONE [28](#)

Numerics

2-WIRE

defined [45](#)

A

about

the TILEncore-Gx36 card [7](#)

this guide [1](#)

accessing product information [4](#)

ACTIVITY LED [29](#)

airflow and ambient temperature requirements [11](#)

B

BIB [45](#)

defined [45](#)

block diagrams

TILEncore-Gx36 [8](#)

Board Information Block

See BIB

Board Test Kit

See BTK

boot options [32](#)

booting

via a PCI [32](#)

booting problems when both USB and PCIe are used [31](#)

BTK [31](#), [35](#)

C

card

operation

verifying [35](#)

resets [21](#)

connecting cables [14](#)

contents in this release [1](#)

D

D1 [28](#), [35](#)

D2 [28](#), [35](#)

D3 [28](#), [35](#)

D4 [28](#), [35](#)

D5 [28](#), [35](#)

D6 [29](#)

D7 [29](#)

DDR3

defined [45](#)

design and layout of the TILEncore-Gx36 card [7](#)

DIP

defined [45](#)

E

ECC SODIMM

defined [45](#)

ECC SODIMMs [25](#)

EEPROM

defined [45](#)

electrical specifications [37](#)

environmental specifications [38](#)

ESD [13](#)

F

features of the TILEncore-Gx36 card [7](#)

functional description [17](#)

G

GbE

defined [45](#)

GBps

defined [45](#)

Gbps

defined [45](#)

GPIO

defined [45](#)

GX READY LED [28](#)

H

handling the TILEncore-Gx36 card [13](#)

hardware requirements [11](#)

host requirements [11](#)

hot swap controller [21](#)

I

I2C

address and device map [19](#)

I2CM0 [19](#)

I2CM1 [19](#)

I2CM2 [19](#)

IDE online help [5](#)

info pages [5](#)

installing
 the TILEncore-Gx card [13](#)
 the TILEncore-Gx card drivers [15](#)
 the TILEncore-Gx36 card [13](#)
intended audience [2](#)

L

LDO [20](#)
LED
 defined [45](#)
 definitions [28, 29](#)
LEDs [27](#)
 D1 [28](#)
 D2 [28](#)
 D3 [28](#)
 D4 [28](#)
 D5 [28](#)
 D6 [29](#)
 D7 [29](#)
 power and health [35](#)
LFM
 defined [45](#)
Linear Tech LTC4211 Hot Swap controller [21](#)
LINK LED [29](#)
low-dropout
 See LDO
LVPECL
 defined [45](#)

M

man pages [5](#)
MDE
 defined [45](#)
 Release 4.0, or later [12](#)
MDE Document Index [5](#)
 location of [5](#)
MDE documentation
 location of [5](#)
mechanical
 dimensions [37](#)
 requirements [11](#)
 specifications [37](#)
Multicore Development Environment
 See MDE

N

notation conventions [3](#)

O

operating system requirements [12](#)
over-temperature protection [20](#)

P

PCI
 booting [32](#)

PCIe
 add-in card [11](#)
 defined [45, 46](#)

POR

 defined [46](#)

power

 and health LED indicators [35](#)
 requirements [12](#)
 specifications [20](#)

POWER IN LED [28](#)

POWER OK LED [28](#)

power-on reset [21](#)

pre-installation guidelines [13](#)

preventing electrostatic discharge
 See ESD

product information [4](#)

Q

QSFP
 defined [46](#)

R

RoHS statement [40](#)

ROM

 defined [46](#)

RSHIM_COORD SPR [3](#)

running programs on the TILEncore-Gx card [31](#)

S

safety requirements [39](#)
shock and vibration specifications [38](#)

SODIMM

 defined [46](#)

software requirements [12](#)

SPD ROM

 defined [46](#)

specifications

 electrical [37](#)
 environmental [38](#)
 mechanical [37](#)
 power [20](#)
 shock and vibration [38](#)

SPI

 defined [46](#)

SPI-SROM [46](#)

SW1 [21](#)

system requirements [11](#)

T

technical and customer support [4](#)

temperature requirements [11](#)

testing the TILEncore-Gx card [35](#)

tile-btk [35](#)

 using against the USB interface [32](#)

tilegxpci driver [31, 32](#)

- tile-monitor [31](#)
- TILEncore-Gx card
 - installing [13](#)
 - drivers [15](#)
 - running programs [31](#)
 - testing [35](#)
- TILEncore-Gx36 block diagram [8](#)
- TILEncore-Gx36 card
 - about [7](#)
 - design and layout [7](#)
 - features [7](#)
 - handling [13](#)
 - installing [13](#)
- TILERA_ROOT
 - defined [4](#)
- tileusb driver [31](#)

- U**
- USB and PCIe usage
 - problems in booting [31](#)
- using tile-btk against the USB interface [32](#)

- V**
- verifying that the card operates properly [35](#)
- voltage
 - monitor/sequencer [21](#)

- W**
- WIP bit [3](#)

- X**
- x8
 - edge connector [11](#)
 - PCIe connector [11](#)
 - PCIe interface [11](#)
- XAUI
 - defined [46](#)

